

RESERVIO

Software Requirements Specification

Requirements Analysis

Lab 05 Deliverable | April 2026 | Klea Hila

1. User Scenarios List (US_17 – US_20)

The following table provides a short description of each new user scenario added in this iteration.

Nr	Name	Description
US_17	Generate Financial Report	Manager: The manager generates a financial report for a selected time period, viewing total revenue, payments received, and outstanding balances.
US_18	Manage Room Inventory	Manager: The manager views all rooms across the property, updates room details (type, price, status), adds new rooms, or deactivates rooms no longer in service.
US_19	Mark Room as Clean	Cleaner: After finishing the cleaning of a room, the cleaner marks the room status as clean in the system so it becomes available for new guests.
US_20	View Assigned Cleaning Tasks	Cleaner: The cleaner views the list of rooms assigned for cleaning, including room number, floor, and current status, in order to plan the cleaning schedule.

2. User Scenarios Extended

Each user scenario is expanded below with the detailed steps required to reach the desired output.

US_17: Generate Financial Report

1. Manager navigates to the Reports section from the dashboard.
2. Manager selects the report type (e.g., Revenue Summary).
3. Manager picks a date range (start date and end date).
4. System queries all completed reservations and payments within the selected period.
5. System calculates total revenue, total payments received, and outstanding balances.
6. System displays the report on screen in a tabular format.
7. Manager optionally exports the report as a PDF or CSV file.
8. If no data exists for the period: System displays a 'No records found' message.

US_18: Manage Room Inventory

1. Manager navigates to the Room Management section.
2. System displays a list of all rooms with their type, price, floor, and current status.
3. Manager selects an action: Add Room, Edit Room, or Deactivate Room.
4. If Adding: Manager fills in the room form (number, type, floor, price, amenities) and submits.
5. If Editing: Manager updates the relevant fields and saves changes.
6. If Deactivating: Manager confirms the action; system sets room status to 'Inactive'.
7. System validates inputs and saves the updated room record.
8. System reflects the changes immediately in the room availability calendar.

US_19: Mark Room as Clean

1. Cleaner logs into the system and navigates to their task list.
2. Cleaner selects the room they have just finished cleaning.
3. System displays the room detail page showing current status as 'Needs Cleaning'.
4. Cleaner clicks 'Mark as Clean'.
5. System updates the room status to 'Clean' and records the timestamp.
6. System notifies the receptionist that the room is ready for check-in.
7. If the cleaner selects the wrong room: System displays a confirmation prompt before saving.

US_20: View Assigned Cleaning Tasks

1. Cleaner logs into the system.
2. System automatically displays the cleaner's task list for the current day.
3. Each task shows: room number, floor, room type, and cleaning priority.
4. Cleaner can filter tasks by status (Pending, In Progress, Done).
5. Cleaner clicks on a task to view detailed room notes left by the receptionist.
6. Cleaner marks a task as 'In Progress' while cleaning.
7. Upon completion, cleaner marks task as 'Done' (triggers US_19 flow).

8. If no tasks are assigned: System displays 'No tasks assigned for today'.

3. Use Cases

Formal use cases for each new user scenario, including actors, preconditions, main and alternative sequences, non-functional requirements, and postconditions.

UC_17: Generate Financial Report	
UC Name	Generate Financial Report
Summary	The manager selects a time period and generates a financial summary report containing revenue, payments, and outstanding balances.
Dependency	US_01 (User logs in), US_02 (Make a reservation), US_08 (Check-Out guest)
Actors	Manager (primary actor)
Preconditions	• Manager is authenticated and has Manager role. • At least one completed reservation exists in the system.
Description of Main Sequence	• Manager navigates to the Reports section. • System displays available report types (Revenue Summary, Occupancy Report). • Manager selects 'Revenue Summary' and specifies a date range. • System validates the date range input. • System queries all completed payments within the selected period. • System calculates total revenue, payments received, and outstanding balances. • System renders the report in a tabular format on screen. • Manager optionally clicks 'Export' to download the report as PDF or CSV. • System generates the file and triggers a browser download.
Description of Alternative Sequence	1. Invalid date range: a. System highlights the date fields in red. b. System displays an error: 'End date must be after start date'. c. Manager corrects the input and resubmits. 2. No data for selected period: a. System displays a 'No records found for the selected period' message. b. Manager adjusts the date range or cancels.
Non-Functional Requirements	• Report generation must complete within 3 seconds for datasets up to 1,000 records. • Exported files must be formatted correctly and open in standard PDF/spreadsheet viewers. • Financial data must be visible only to Manager role; no other roles can access this page.
Postconditions	• Financial report is displayed on screen or downloaded as a file. • No data is modified as a result of this use case.

UC_18: Manage Room Inventory	
UC Name	Manage Room Inventory
Summary	The manager views, adds, edits, or deactivates hotel rooms to maintain the accurate room inventory used by the reservation system.
Dependency	US_01 (User logs in)
Actors	Manager (primary actor)

Preconditions	• Manager is authenticated and has Manager role. • The Room Management section is accessible from the dashboard.
Description of Main Sequence	• Manager navigates to the Room Management section. • System displays a paginated list of all rooms with number, type, floor, price, and status. • Manager selects 'Add New Room' or clicks an existing room to edit. • System displays the room form pre-filled with existing data (or blank for new). • Manager fills in or updates: room number, type, floor, price per night, amenities. • Manager clicks 'Save'. • System validates all required fields and checks for duplicate room numbers. • System saves the record and refreshes the room list. • System confirms with a success message: 'Room saved successfully'.
Description of Alternative Sequence	1. Duplicate room number: a. System displays: 'Room number already exists'. b. Manager changes the room number and resubmits. 2. Deactivate room: a. Manager clicks 'Deactivate' on an existing room. b. System shows confirmation modal: 'Are you sure? Active reservations may be affected'. c. Manager confirms; system sets room status to Inactive. d. Room no longer appears in availability searches. 3. Missing required field: a. System highlights empty required fields in red. b. Manager fills in missing data and resubmits.
Non-Functional Requirements	• Room list must load within 2 seconds for properties with up to 500 rooms. • Changes to room price must not retroactively affect confirmed reservations. • Only Manager role may add, edit, or deactivate rooms.
Postconditions	• Room record is created or updated in the database. • Changes are immediately reflected in the availability calendar and reservation flow.

UC_19: Mark Room as Clean	
UC Name	Mark Room as Clean
Summary	After completing room cleaning, the cleaner updates the room status to 'Clean' so the room becomes available for guest check-in.
Dependency	US_01 (User logs in), US_20 (View Assigned Cleaning Tasks)
Actors	Cleaner (primary actor), System (notifies Receptionist)
Preconditions	• Cleaner is authenticated and has Cleaner role. • The room's current status is 'Needs Cleaning'.
Description of Main Sequence	• Cleaner opens the task list and selects a room with status 'Needs Cleaning'. • System displays the room detail page with room notes and cleaning instructions. • Cleaner clicks 'Mark as Clean'. • System displays a confirmation prompt: 'Confirm room [number] is clean?'.

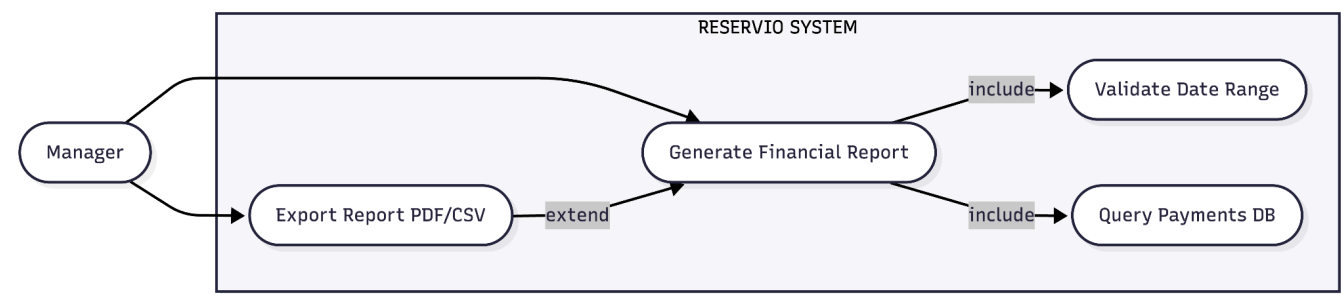
	- Cleaner confirms. - System updates the room status to 'Clean' and records the completion timestamp and cleaner ID. - System sends a real-time notification to the receptionist dashboard. - System marks the cleaner's task as 'Done'.
Description of Alternative Sequence	1. Cleaner selects wrong room: - System displays confirmation prompt before applying the change. - Cleaner cancels and selects the correct room. 2. Room already marked clean: - System displays: 'This room is already marked as clean'. - No changes are made.
Non-Functional Requirements	- Status update must be reflected in the receptionist's dashboard within 2 seconds (real-time via WebSocket). - Timestamp and cleaner ID must be stored immutably for audit purposes. - Only users with Cleaner role may trigger this status update.
Postconditions	- Room status is updated to 'Clean' in the database. - Timestamp and cleaner ID are recorded for the cleaning event. - Receptionist receives a real-time notification that the room is ready.

UC_20: View Assigned Cleaning Tasks	
UC Name	View Assigned Cleaning Tasks
Summary	The cleaner views their personalised task list showing which rooms require cleaning, along with priority, floor, and any notes from the receptionist.
Dependency	US_01 (User logs in), US_08 (Check-Out guest)
Actors	Cleaner (primary actor)
Preconditions	- Cleaner is authenticated and has Cleaner role. - At least one room has been assigned for cleaning (triggered by check-out).
Description of Main Sequence	- Cleaner logs in; system automatically loads the task list for the current day. - System displays tasks sorted by priority: urgent rooms first, then by floor. - Each task card shows: room number, floor, room type, status (Pending / In Progress / Done), and any receptionist notes. - Cleaner filters or searches tasks by status or floor. - Cleaner taps a task to view detailed notes. - Cleaner marks a task as 'In Progress' to indicate cleaning has started. - Upon completion, cleaner marks task as 'Done' (initiates UC_19).
Description of Alternative Sequence	1. No tasks assigned: - System displays: 'No cleaning tasks assigned for today' - Cleaner can contact the manager or receptionist. 2. New task assigned in real time: - System receives a WebSocket push and adds the new task to the list without page refresh. - Cleaner sees the task appear instantly.
Non-Functional Requirements	- Task list must load within 1.5 seconds on mobile devices at 375px viewport. - Real-time task updates must be delivered within 2 seconds via WebSocket.

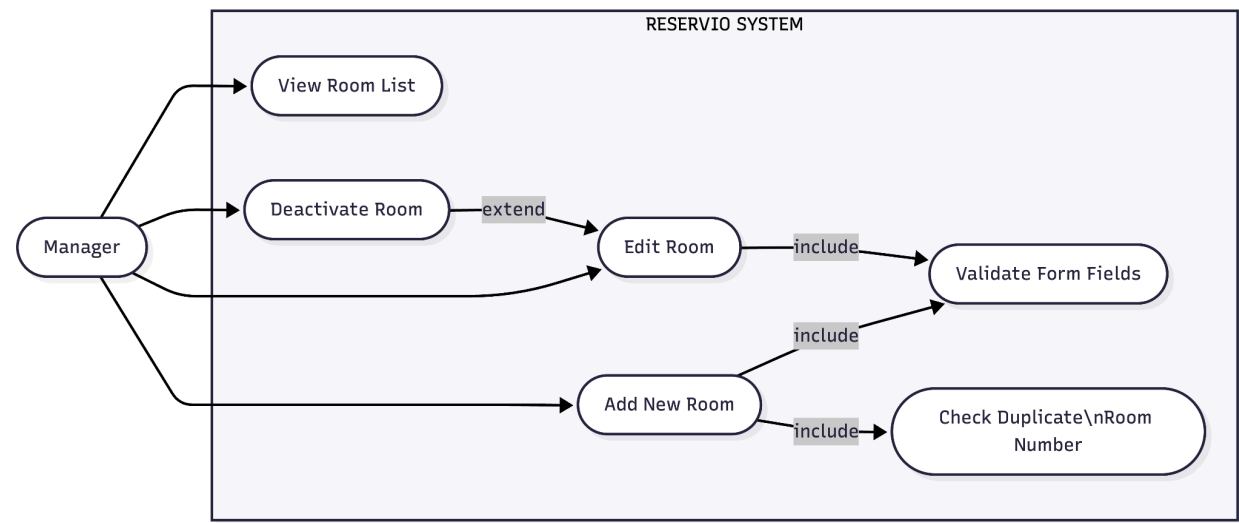
	Only tasks assigned to the logged-in cleaner are visible; cross-cleaner data is not accessible.
Postconditions	- Cleaner has a clear view of all assigned tasks and their priorities. - Task statuses are updated in real time for receptionist and manager visibility.

4. Use Case Diagrams

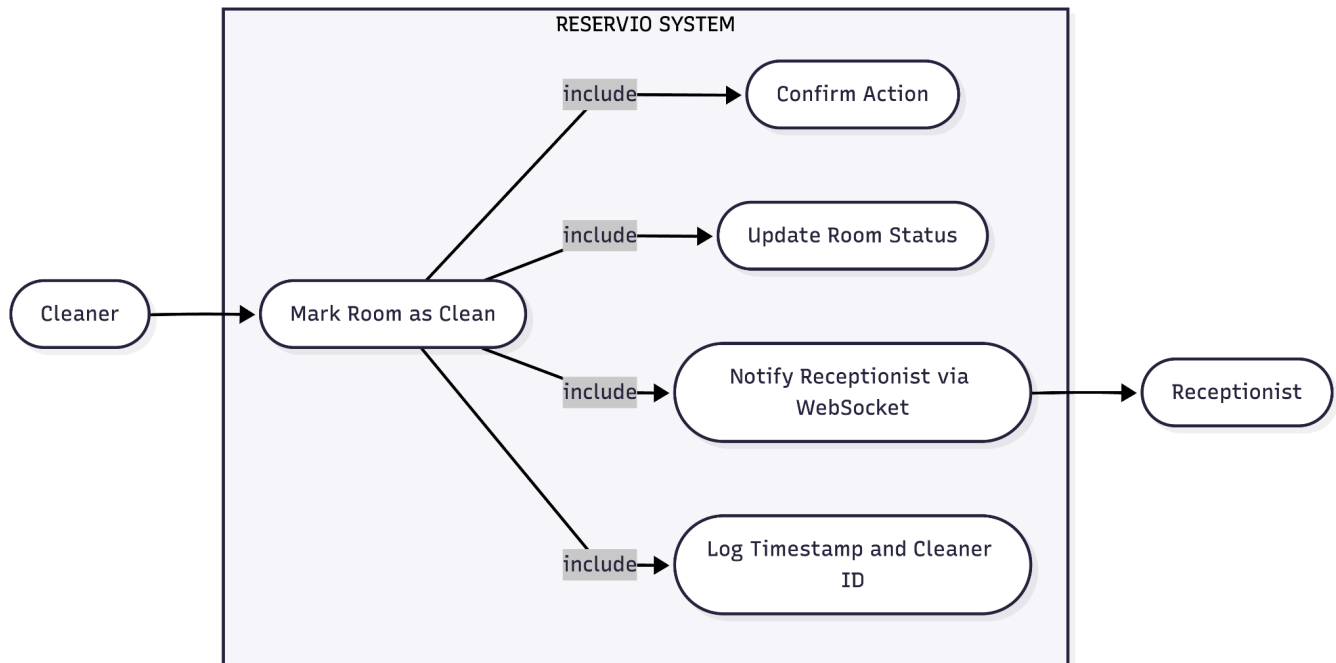
UC_17



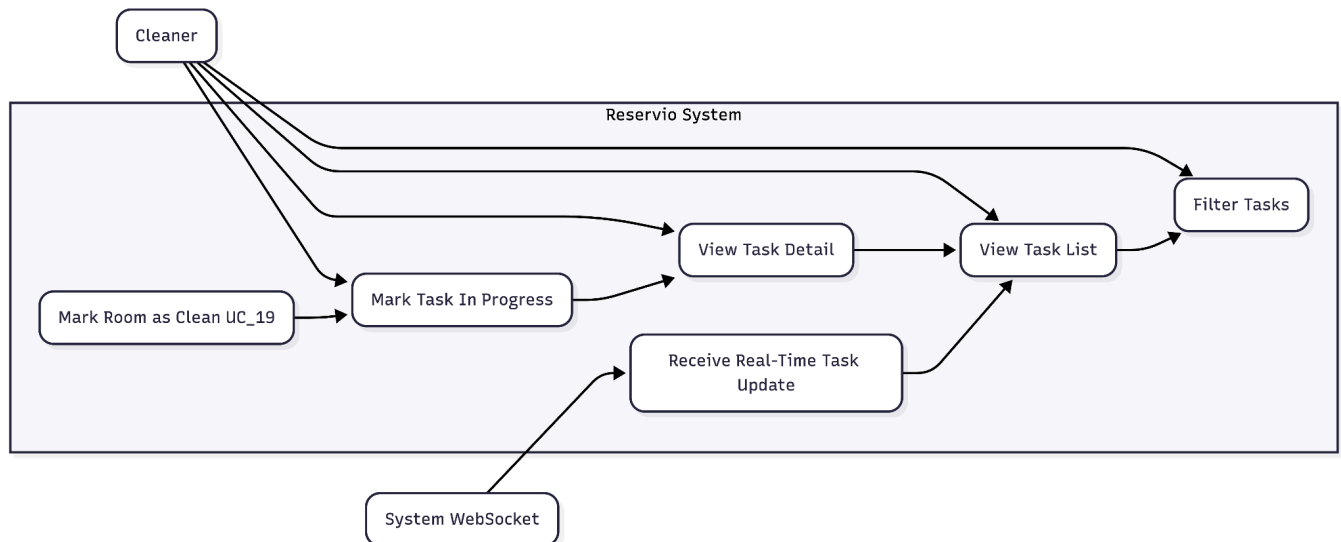
UC_18



UC_19



UC_20



5. Functional Requirements

The following functional requirements are derived from the use cases above and cover all user roles: Guest, Receptionist, Manager, Cleaner, and Super Admin.

Req#	Requirement	Comments
FR_01	The system shall allow all user roles (Guest, Receptionist, Manager, Cleaner) to log in using a unique username and password.	Credentials are validated against hashed passwords stored in the database.
FR_02	The system shall grant role-based access upon successful login, routing each user to their respective dashboard.	Roles: Guest, Receptionist, Manager, Cleaner, Super Admin.
FR_03	The system shall allow a Guest to select room type, check-in date, and check-out date to make a reservation.	System checks real-time room availability before confirming.
FR_04	The system shall generate an invoice automatically when a reservation is confirmed.	Invoice includes room type, dates, total price, and reservation ID.
FR_05	The system shall allow a Guest to view all current and past reservations linked to their account.	Past reservations display final invoice and feedback status.
FR_06	The system shall allow a Guest to cancel a reservation with status 'Pending' or 'Confirmed' prior to check-in.	Cancelled reservations are flagged and removed from room availability.
FR_07	The system shall allow a Guest to submit a feedback form after their reservation status is set to 'Completed'.	Feedback is linked to the reservation and notifies the manager.
FR_08	The system shall allow a Guest to order extra services (e.g., room service, laundry) during an active stay.	Extra services are logged and added to the guest's invoice.
FR_09	The system shall allow a Receptionist to check in a guest by searching their reservation and assigning an available room.	Check-in changes room status to 'Occupied' and reservation status to 'Active'.
FR_10	The system shall allow a Receptionist to check out a guest, finalize their invoice, and mark the room as 'Needs Cleaning'.	Check-out triggers a cleaning task assigned to available cleaners.
FR_11	The system shall allow a Manager to add a new staff member by filling in a form with name, role, email, and temporary password.	System sends the new staff member a login invitation email.
FR_12	The system shall allow a Manager to deactivate a staff member account, immediately revoking their system access.	Deactivated accounts cannot log in; historical data is preserved.
FR_13	The system shall allow a Manager to generate a financial report for a selected date range, showing total revenue and outstanding balances.	Report data is sourced from completed reservations and payments.
FR_14	The system shall allow a Manager to add, edit, or deactivate hotel rooms, including setting room type, floor, price, and amenities.	Changes to room price do not retroactively affect confirmed reservations.
FR_15	The system shall allow a Manager to view and respond to guest feedback submitted after completed stays.	Manager responses are stored and optionally visible to the guest.

Req#	Requirement	Comments
FR_16	The system shall display a list of assigned cleaning tasks to the Cleaner upon login, sorted by priority and floor.	Tasks are generated automatically upon guest check-out.
FR_17	The system shall allow a Cleaner to update a room status to 'Clean' after completing cleaning, with timestamp and cleaner ID recorded.	Status change triggers a real-time notification to the receptionist dashboard.
FR_18	The system shall allow a Cleaner to mark a cleaning task as 'In Progress' before updating it to 'Done'.	In-Progress status is visible to Manager and Receptionist in real time.
FR_19	The system shall support multi-tenancy, ensuring each hotel's data (guests, rooms, reservations, staff) is fully isolated from other tenants.	Enforced via tenantId on every database table and all API queries.
FR_20	The system shall apply dynamic pricing rules nightly, adjusting room base prices according to occupancy thresholds defined by the Manager.	Nightly cron job evaluates pricing; manual override by Manager is also permitted.

6. Non-Functional Requirements

Non-functional requirements define the quality attributes, constraints, and standards the Reservio system must adhere to across all user roles and features.

i) Product Requirements

User Interface Requirements

Sub-Category	Requirement
Responsive Design	All guest-facing pages must be fully functional on mobile browsers at a minimum viewport width of 375px (Tailwind CSS mobile-first).
Consistent Styling	The UI must follow the Reservio design system (colours, typography, spacing) across all roles and pages.
Role-Based Views	Each role (Guest, Receptionist, Manager, Cleaner) must see only the navigation and features relevant to their role; no cross-role UI leakage.
Error Feedback	All form validation errors must be displayed inline next to the relevant field within 500ms of form submission.

Learnability

Sub-Category	Requirement
Onboarding	New staff users must be able to complete their primary workflow (check-in, mark clean, etc.) without training documentation, within 10 minutes of first login.
Tooltips & Labels	All action buttons and icons must include descriptive labels or tooltips that describe the action in plain language.

Accessibility

Sub-Category	Requirement
WCAG Compliance	Guest-facing pages must meet WCAG 2.1 Level AA standards, including sufficient colour contrast (4.5:1 minimum ratio) and keyboard navigability.
Screen Reader Support	All interactive elements must have appropriate ARIA labels for screen reader compatibility.

Efficiency / Performance

Sub-Category	Requirement
Page Load Time	All dashboard pages must load within 2 seconds on a standard broadband connection (10 Mbps).
Search Response	Room availability searches must return results within 1.5 seconds for datasets of up to 500 rooms.
Report Generation	Financial reports must generate within 3 seconds for datasets up to 1,000 reservations.

Dependability / Memorability

Sub-Category	Requirement
Session Persistence	User sessions must remain active for up to 8 hours without requiring re-login, using JWT with appropriate expiry.
Graceful Degradation	If WebSocket connection drops, the system must fall back to polling every 10 seconds and notify the user of the connectivity issue.

Errors

Sub-Category	Requirement
Error Messages	All user-facing error messages must be human-readable, describing the problem and suggesting a resolution. No raw stack traces must be exposed.
Data Validation	All form inputs must be validated on both client-side (immediate feedback) and server-side (authoritative check) before any database write.

Security

Sub-Category	Requirement
Password Storage	All passwords must be hashed using bcrypt with a minimum cost factor of 12 before being stored in the database.
JWT Security	JWT tokens must have a maximum expiry of 8 hours. Refresh tokens, if used, must be stored securely and invalidated on logout.
Input Sanitisation	All user inputs must be sanitised server-side to prevent SQL injection and XSS attacks. Parameterised queries via Prisma ORM are mandatory.

ii) Organisational Requirements

Availability

Sub-Category	Requirement
Uptime	The system must target 99% uptime during the academic demonstration period (March–June 2026), subject to Railway free-tier infrastructure limits.
Scheduled Downtime	Any planned maintenance must be performed between 02:00–04:00 local time and communicated to hotel administrators at least 24 hours in advance.

Latency

Sub-Category	Requirement
Critical Operations	Room booking, check-in, and payment processing responses must complete within 2 seconds under normal load.
Real-Time Notifications	WebSocket-delivered notifications (room status updates, new task assignments) must reach clients within 2 seconds of the triggering event.
API Response	All REST API endpoints must respond within 500ms for read operations and within 1 second for write operations under standard load.

Monitoring

Sub-Category	Requirement
Error Logging	All unhandled server errors must be logged with timestamp, endpoint, and stack trace. Logs must be accessible for at least 30 days.
Audit Trail	All room status changes, reservation modifications, and staff management actions must be logged with actor ID and timestamp.

Maintenance

Sub-Category	Requirement
Software Updates	The codebase must follow a CI/CD pipeline (GitHub Actions) that runs ESLint and Jest tests before every deployment to the main branch.
Database Backups	The PostgreSQL database must be backed up daily via Railway's managed backup feature; backups must be retained for a minimum of 7 days.

Operations

Sub-Category	Requirement
Multi-Tenancy	The system must enforce tenant isolation at the database level (tenantId on all tables) and at the API level (middleware tenant-scoping) without exception.
Concurrency	The system must support at least 50 concurrent authenticated users per tenant without performance degradation.

Standards Compliance

Sub-Category	Requirement
Code Standards	All JavaScript code must pass ESLint with the project's configured ruleset (Airbnb or equivalent) before merging to main.
API Design	All backend endpoints must follow RESTful conventions (resource-based URLs, correct HTTP verbs and status codes).

Portability

Sub-Category	Requirement
Browser Support	The application must function correctly on the latest two major versions of Chrome, Firefox, Safari, and Edge without browser plugins.
Device Support	The guest-facing interface must be fully usable on both desktop (1280px+) and mobile (375px minimum) viewports.

iii) External Requirements

Protection (Data Privacy)

Sub-Category	Requirement
Personal Data Handling	Guest personal data (name, email, payment details) must be stored only for the duration of active business relationship and must not be shared with other tenants.
Data Minimisation	The system must collect only the minimum personal data necessary to complete a reservation. No unnecessary PII fields may be added without justification.

Authorization & Authentication

Sub-Category	Requirement
Role Enforcement	Every API endpoint must enforce role-based access control (RBAC); unauthenticated or unauthorised requests must receive HTTP 401 or 403 respectively.
Stateless Auth	Authentication must be implemented using JSON Web Tokens (JWT); no server-side session store may be used.
Super Admin Isolation	The Super Admin role must only be accessible via admin.reservio.com and must not be reachable from any hotel-specific tenant subdomain.

Legislative Requirements

Sub-Category	Requirement
GDPR Awareness	As the system stores EU personal data, it must implement mechanisms for data subject requests (access, deletion) in compliance with GDPR principles, even if full GDPR certification is not in scope for this academic project.
No Real Payments	No actual payment card data may be stored or processed; payment processing is explicitly out of scope and must never be implemented without a certified payment gateway.